Two barriers to equitable access: WCAG conformance and pre-consent tracking on Ecuadorian and top-ranked university websites

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Abstract

Purpose. The institutional website is the channel through which applicants find programmes, apply and enrol, so a barrier embedded in it restricts access to the institution itself. This article measures two such barriers on the same pages, namely conformance with the Web Content Accessibility Guidelines (WCAG) and the tracking that occurs before any consent is given, and asks whose location determines the protection a visitor receives.

Methods. The home page and first-level privacy notice of 126 universities were audited under one protocol in August 2026: the complete census of the 63 Ecuadorian higher education institutions and a benchmark of 63 institutions drawn from the consensus of three international rankings. Five documentary indicators were coded and three live indicators recorded with a headless browser and axe-core 4.13. The live audit was then replicated from four vantage points, Ecuador, Germany, the United Kingdom and the United States, with three passes at each.

Results. Ecuadorian institutions published less: 61.9% a privacy notice and 38.1% a privacy contact, compared with 93.7% and 85.7% in the benchmark group. Both groups failed the accessibility checks, but Ecuadorian failures concentrated at the basic tier, 45% of failing nodes at level A compared with 19%. Pre-consent tracking did not differ between the groups, and the design had no power to establish equivalence. It did differ with the visitor's location: on 15 August 2026 the same pages set tracking cookies for 67.8% of visits from Ecuador and 60.3% of visits from Germany (Cochran $\boldsymbol{Q}$, $\boldsymbol{p = 0.0012}$; Ecuador–Germany Holm-adjusted $\boldsymbol{p = 0.023}$). The pattern is attributable to the advertising vendors embedded in the pages rather than to the institutions: Microsoft, at 14 sites, and TapAd, at 4, set no cookies at the German and British vantage points while setting them at the Ecuadorian and United States ones, on Ecuadorian, Chinese and United States university sites as readily as on European ones.

Conclusion. Ecuador's distinctive shortfalls are in what institutions publish and in basic-level accessibility, and both are remediable without new legislation. Pre-consent exposure, by contrast, is delimited by country lists that advertising vendors operate, which are neither published, nor harmonised, nor supervised, and the population outside those lists is the international applicant pool of Latin America, Africa and much of Asia.

Keywords: web accessibility, WCAG conformance, personal data protection, cookie consent, geo-differential tracking, university websites, Ecuador

1 Introduction

The institutional website is the channel through which a prospective student finds programme information, submits an application, completes enrolment and reaches library, tutoring and administrative services. A barrier embedded in that channel is therefore not a cosmetic defect but a restriction on access to the institution. Universal access research has long framed this as a design problem rather than a remediation problem: environments and services should be usable by the widest possible range of people from the outset, and accessibility standards are the operational expression of that principle for digital products [29]. The Web Content Accessibility Guidelines (WCAG), currently in version 2.2 [67], provide the testable criteria, and several legal regimes make them binding, among them the European Accessibility Act [22] and, in the United States, the Americans with Disabilities Act (ADA) and Section 508 of the Rehabilitation Act [62, 63].

Personal data protection is a second barrier operating on the same channel and, we argue, on the same principle. A visitor who must accept unexplained tracking in order to consult a study programme, or who cannot find out who processes their application data or how to object, faces a cost of participation that is unevenly distributed: it falls hardest on users with least legal literacy, least technical means to refuse and least ability to seek an alternative provider. Read this way, opaque privacy governance and inaccessible interfaces are two mechanisms of the same phenomenon, namely the unequal cost of using a public-facing service. This framing motivates studying them together rather than as two separate compliance checklists.

Empirical work has treated the two literatures separately. On accessibility, audits of university portals report widespread barriers in the highest-ranked universities assessed against WCAG 2.2 [33], in European institutions [31], in the Gulf region [24] and in public online services generally [4, 30], with a long-documented lag in Latin America [2, 9]. On privacy, large-scale measurement has established that written rules and observed behaviour advance at different speeds: the General Data Protection Regulation (GDPR) [21] multiplied notices and banners without proportionate change in tracking [13, 32, 60], consent interfaces frequently fail the legal conditions for valid consent [27, 28, 40, 47], cookie managers often do not block what they claim to block [7, 56], and in higher education specifically, policies have been observed to change to look compliant rather than to be compliant [42]. To our knowledge, no study has measured both barriers on the same set of institutional sites, with the same protocol and on the same date.

Ecuador is an informative setting for that measurement. Its Organic Law on the Protection of Personal Data (Ley Orgánica de Protección de Datos Personales, LOPDP) entered into force in 2021 and was implemented by regulation only in 2023 [5, 53], placing it in the most recent wave of a regional diffusion process that comparative work has shown to be uneven in commitment and enforcement [10, 11]. The country also has a closed, enumerable population of 63 recognised higher education institutions (HEIs), which allows a complete census rather than a sample.

This article addresses five research questions.

- **RQ1.** How do Ecuadorian HEIs and a top-ranked international benchmark group differ in the privacy governance visible on their institutional home page, namely a published privacy notice, a cited legal framework, an enumeration of data-subject rights and an identified privacy contact?
- **RQ2.** How do the two groups differ in observed pre-consent behaviour, that is, in the cookies set before any interaction with a consent interface, when both are observed from the same vantage point outside the European Union?
- **RQ3.** How do the two groups differ in automatically detectable WCAG conformance, and does the difference lie in the volume of failures or in the conformance level at which they occur?
- **RQ4.** Within each group, is measured accessibility associated with observed restraint in pre-consent tracking?

- **RQ5.** Does the pre-consent behaviour of the same page depend on where the *visitor* is located, and if so, does the boundary follow the law that binds the institution or something else?

The contributions are the following. First, a paired measurement of accessibility and pre-consent tracking on 126 university home pages with a single, fully specified and reproducible protocol, released with its configuration, its tracking-cookie taxonomy and its raw output. Second, an inferential reanalysis that separates the differences the design can support from those it cannot, including an explicit equivalence test and a power statement for the null results the study reports. Third, a four-vantage replication of the live audit, from Ecuador, Germany, the United Kingdom and the United States, which converts a legal conjecture into a measurement: the protection a visitor receives is shown to depend on the visitor's location and to be enforced by advertising vendors rather than by the institutions or by the law that binds them. Fourth, an indicator-by-jurisdiction mapping that separates observed conduct from legal non-compliance, since most of the eleven jurisdictions studied impose no general prior-consent duty for cookies.

The remainder of the article is organised as follows. Section 2 reviews the two literatures and states the gap. Section 3 describes the design, the sampling, the codebook, the audit protocol and the statistical analysis. Section 4 reports the results for each research question. Section 5 interprets them, derives implications for practice and policy, and states the threats to validity. Section 6 concludes and sets out the future work the present design cannot deliver.

2 Related work

2.1 Web accessibility as a condition of equitable access

Universal design treats accessibility as a property to be built into a service rather than added to it. The tradition is not, however, terminologically settled, and the lack of settlement has methodological consequences: Persson et al [52] trace the parallel histories of universal design, inclusive design, accessible design and design for all and show that the absence of a shared definition weakens both measurability and conformance assessment, which is why the present article fixes its operational definitions before reporting any result. Two commitments of that tradition bear directly on this study. The first is that a barrier is a property of the encounter between a person and a service rather than of the person, so an audit describes what a service does to the range of people obliged to use it. The second is distributive: Abascal et al [1] argue that universal accessibility formulated without regard to socioeconomic and geopolitical context reaches only a fraction of the population it claims to serve, and that exclusion by location must be addressed alongside exclusion by ability. That argument is the one Section 5.2 returns to, because the boundary this study measures turns out to be geographical rather than functional. On the instrument side, Etxabe-Antia et al [17] review 596 accessibility recommendations across digital technologies and find that established web technologies converge on WCAG while emerging ones do not, which supports the choice of WCAG as the reference standard for institutional web pages; and media accessibility research has examined how the same principle interacts with ecological and context-of-use considerations [29]. Applied to higher education, the systematic review by Campoverde-Molina et al [9] covers university website accessibility worldwide and reports persistent non-conformance across regions, methods and time periods. More recent single-study evidence is consistent with that picture: Krittika and Shimray [33] evaluate the top 25 universities of the Quacquarelli Symonds (QS) world ranking and the top 25 of its Indian ranking against WCAG 2.2 and find failures in both sets; Krejtz et al [31] report on the accessibility information published by European universities; and Fakrudeen [24] reach similar conclusions for the Gulf region. Outside higher education, Aljedaani [4] and Iniesto and Rodrigo [30] document the same dominant error types on public-sector portals, principally insufficient contrast and links without an accessible name. In Latin America, Acosta-Vargas et al [2] and Acosta-Vargas et al [3] provide comparative evidence of a regional lag, and Maldonado-Garcés et al [39] extend the argument from the digital to the physical campus in Ecuador.

Two methodological cautions follow from this literature and shape the present design. First, automated tools demonstrate non-conformance but cannot certify conformance: Fischer et al [25] quantify the coverage of automated checkers and show that they address only a minority of the WCAG success criteria, so an automated pass is not evidence of an accessible site. Second, results depend on the tool, on the rule set enabled and on the rendering conditions, which is why this study reports its full configuration rather than only its tool name.

2.2 Privacy governance and web privacy measurement

The measurement literature on web privacy provides both the instruments and the cautionary findings for the present study. Libert [37] and Englehardt and Narayanan [16] established large-scale browser-based measurement of third-party tracking and showed that the phenomenon is concentrated, persistent and largely invisible to users. After the GDPR became applicable, Degeling et al [13] and Trevisan et al [60] found that notices and banners proliferated without a proportionate reduction in tracking, and Kretschmer et al [32] reached comparable conclusions over a longer window. A second strand shows that the interfaces themselves are frequently non-compliant: Matte et al [40] found that a large share of banners built on the Interactive Advertising Bureau (IAB) Europe Transparency and Consent Framework registered a consent signal inconsistent with the user's choice; Nouwens et al [47], Gray et al [27] and Habib et al [28] document design patterns that make acceptance the path of least resistance; and Sanchez-Rola et al [56] and Bollinger et al [7] show that consent-management platforms often fail to block the trackers they purport to gate. On the textual side, privacy policies remain long and hard to read [6, 23].

A third strand, directly relevant here, treats the location of the observer as an experimental variable rather than as a fixed property of the study. Dabrowski et al [12] crawled the same sites from clients in the European Union and in the United States and found that 26.6% of the cookie-setting sites placed persistent cookies for the United States client and none for the European one, an asymmetry that was almost absent in the opposite direction. van Eijk et al [15] crawled from eighteen vantage points and reached a more conservative conclusion, that the top-level domain of the site explains more of the variation in cookie notices than the location of the visitor does, with `.com` sites the exception. Rasaii et al [55] confirm the pattern on the notice side, reporting consent banners 56% more prevalent when a site is visited from within the European Union. The three studies agree that where a measurement is taken from changes what is measured; they do not agree on what governs the difference.

In higher education, Liu and Khalil [38] review privacy issues in learning analytics and Mutimukwe et al [43] examine online proctoring, both concerned with internal systems rather than the public portal. Closest to the present work, Munot et al [42] analyse how universities reshaped their privacy policies after the GDPR and distinguish compliance achieved by design from compliance achieved by appearance.

2.3 The gap addressed here

Three gaps follow from the two literatures reviewed above, and the design of this study is a response to each of them.

The first is a gap of scope. The two literatures share a subject, the institutional website, and an underlying concern, the cost that a service imposes on the people who must use it, but they have not been brought together on a common set of sites. Accessibility studies do not measure tracking; privacy measurement studies do not audit conformance. Because the two are always measured apart, a question that matters for practice cannot be answered at all: whether an institution that commits to one form of access also commits to the other, or whether the two are independent properties that a single institutional commitment does not deliver. Section 4.5 tests that association directly, which requires both barriers to be measured on the same sites, on the same day, with one protocol.

The second is a gap of vantage point. The geo-differential literature is built almost entirely on an axis between the European Union and the United States, and it attributes what it finds to the site: a site either discriminates by visitor location or it does not. Two consequences follow. Latin America appears in none of these designs, so what a visitor located under a young data-protection regime actually receives has not been measured;

and because the unit of attribution is the site, the mechanism is left unresolved, since an asymmetry observed at the site cannot distinguish a decision taken by the operator of the site from one taken by the third parties whose code the site embeds. This distinction is not academic: it determines whether a remedy addressed to universities can work at all. Section 4.6 measures from four vantage points, one of them Ecuadorian, and attributes the difference to named third-party domains rather than to the sites that host them.

The third is a gap of population. Studies of university websites, in both literatures, rest overwhelmingly on ranked or convenience samples, which by construction describe institutions selected for visibility. A complete national population supports a different claim, because it carries no sampling error and no selection on prominence, and because it is the population that a national regulator is actually obliged to supervise. Setting such a population against an elite benchmark under an identical protocol is what allows the differences reported in Section 4 to be read as differences between the two groups rather than as differences between two studies. That combination, two barriers, four vantage points and a complete national population against a benchmark, is what this article contributes.

3 Methods

3.1 Design and unit of analysis

The study is a cross-sectional, two-group comparison carried out in August 2026. The unit of analysis is the *institutional home page and the first-level privacy notice linked from it.* This is a deliberately narrow unit and it is stated explicitly because it determines what the results can mean: a document that exists on a faculty subdomain but is not reachable from the home page is coded as absent, since a prospective applicant arriving at the institutional portal would not find it. Where a first-level notice referred the reader onward to a more detailed statement, the referenced statement was read and coded, but only when it was reachable in one click from the first-level notice.

The comparison is a benchmarking exercise against an aspirational ceiling, not an estimate of a country effect. The benchmark group is not a sample of "the rest of the world": it is the extreme upper tail of an international ranking distribution, and it differs from the Ecuadorian population in budget, technical staffing and regulatory maturity as well as in country. Every difference reported in Section 4 is therefore open to interpretation as an effect of institutional resources rather than of jurisdiction, and no causal attribution to country is made anywhere in this article.

3.2 Sampling

The Ecuadorian group is a complete census of the 63 universities and polytechnic institutions recognised by the national higher-education authorities, the Secretariat of Higher Education, Science, Technology and Innovation (Secretaría de Educación Superior, Ciencia, Tecnología e Innovación, SENESCYT) and the Council of Higher Education (Consejo de Educación Superior, CES), as listed on 11 August 2026. Institutions created in 2024 and 2025 are included, and are identified as such in Appendix D so that readers can exclude them if they consider the comparison unfair to institutions that have not yet completed their first academic cycle.

The benchmark group was constructed by consensus of three rankings in their most recent editions: the QS World University Rankings 2026, the Times Higher Education (THE) World University Rankings 2026 and the Academic Ranking of World Universities (ARWU) 2025 [54, 57, 59]. From each ranking the top 75 was taken; each institution received the mean of its three positions, with absence from a ranking's top 75 imputed as position 76; the number of rankings in which an institution appeared took precedence over that mean, so the 46 institutions present in all three head the list and the remaining 17 were drawn, by mean position, from those present in two. The cut-off institution was the University of Queensland and the highest-placed institution excluded was the London School of Economics. The group size was set at 63 to match the Ecuadorian census. Figure E1 in Appendix E shows the resulting distribution by country; the United States contributes 24 institutions, the United Kingdom 7, China 6 and Australia 5, with the remainder spread over eleven further jurisdictions.

The aggregation rule involves three arbitrary choices, namely the imputation constant, the depth of the ranking cut and the lexicographic priority given to ranking coverage. These were not varied, and the sensitivity of the composition to them is reported as a limitation in Section 5.5 rather than claimed as robust.

3.3 Dimensions, operational definitions and codebook

Five documentary dimensions and three live dimensions were coded. The operational definitions are given in full in Appendix A; the essential points are the following.

Privacy notice. Coded as present when a document whose stated purpose is to describe the processing of personal data is reachable from the home page. Coded **P** (partial) when the document exists but its stated scope covers only part of the institution's processing, for example admissions only, postgraduate programmes only or a mobile application only. Coded NV (not verifiable) when the site could not be retrieved or the document was not machine-readable.

Applicable legal framework cited. Coded as present when the notice names a specific legal instrument. Generic formulations such as "applicable law" or "data protection legislation" were coded as absent. In Ecuador the relevant instrument is the LOPDP and its implementing regulation [5, 53]; elsewhere the framework applicable to the institution was accepted, as set out in the indicator-by-jurisdiction mapping of Appendix B.

Data-subject rights listed. Coded as present when the notice enumerates at least three of the rights of access, rectification, erasure, restriction, objection and portability, together with a route to exercise them. A generic mention of "your rights" without enumeration was coded as absent.

Privacy contact or data protection officer. Coded as present when the notice gives a named role or a dedicated address or form for privacy matters. A general institutional contact form was coded as absent. We distinguish throughout between the *controller*, which is the institution itself and is therefore never absent, and the *data protection officer (DPO) or privacy contact*, which is the designated route through which a data subject can act.

Transport security. The presence of a valid Transport Layer Security (TLS) certificate was recorded. This indicator is reported under the heading of transport security and not of data protection: it measures confidentiality in transit and says nothing about what the site does with the data once received. It was verified separately from the automated audit, since the audit browser was configured to ignore certificate errors in order to reach sites with faulty chains.

Live dimensions. Pre-consent cookies, tracking cookies, consent banner and consent-management platform (CMP) were recorded by the automated audit described in Section 3.5, as was WCAG conformance.

3.4 Documentary coding and its verification

The documentary dimensions were coded by the authors from the rendered documents. The study does not report an inter-coder reliability coefficient, because double coding was not performed; this is a material limitation and is stated as such in Section 5.5.

To obtain a bounded indication of coding error, a targeted verification sub-study was carried out on 14 August 2026. Six institutions of the benchmark group, selected purposively as those whose individual coding carried the greatest weight in the analysis, were re-examined manually, cell by cell, against the live site: 24 cells in all. Two discrepancies were found. One institution had been coded as publishing no privacy notice when a notice is in fact linked from the home-page footer, and the cell was corrected to present. One institution had been coded as citing an applicable framework when its notice names no instrument and uses only generic formulations, and the cell was corrected to absent. Both corrections are incorporated in every figure and table of this article, and the two cells are marked in Appendix C. The sub-study also revealed a systematic effect of the depth-one unit of analysis: at two further institutions, rights and framework are stated one click below the first-level notice and are therefore recorded as absent under our definition even though the content exists. This is a property of the measure, not an error, but it means that the governance indicators reported here should be read as "visible at the first level" rather than as "published anywhere by the institution".

Because the six institutions were selected purposively rather than at random, two errors in 24 cells is not a reliability estimate and is neither an upper nor a lower bound on the error rate of the full matrix; it is reported only to show that the rate is not negligible. Full double coding of at least 30% of the sample, with a Cohen's κ per indicator and reconciliation of every discrepancy, is the first item of the future work set out in Section 6. The resulting asymmetry, between the reliability evidence available for the documentary indicators and that available for the live ones, is stated as a limitation in Section 5.5.

3.5 Automated audit protocol

Each of the 126 sites was visited once, in August 2026, from a residential connection in Ecuador, with a read-only script. The full configuration is given here so that the measurement can be repeated.

- *Browser and driver.* Chromium driven by Playwright [41], launched headless with the flags `--no-sandbox` and `--disable-dev-shm-usage`. A fresh browser context was created per site, so no state carried over between sites.
- *Viewport and user agent.* Playwright's default desktop viewport of 1280 × 720 CSS pixels was used, with the user-agent string of Chrome 128 on Windows 10 (64-bit). The viewport is reported because it determines the outcome of the contrast, reflow and target-size criteria.
- *Navigation.* `page.goto` with `waitUntil: 'domcontentloaded'`, a 45 s timeout and one retry on failure, followed by a fixed 4 s wait to allow deferred scripts and consent platforms to load. The context was created with `ignoreHTTPSErrors: true` so that sites with certificate problems could still be audited. No scrolling and no interaction with any banner were performed at any point.
- *Cookie state.* The full cookie jar of the context was read before any interaction. A cookie was classified as *tracking* when its name matched a declared allow-list of analytics, advertising and session-replay families, reproduced in full in Appendix A. The list covers Google Analytics and Google Ads, Meta, TikTok, Microsoft Clarity and Bing, Hotjar, Matomo, Siteimprove, Pinterest, Adobe, Tealium, Baidu, Quantcast, LinkedIn, X and the YouTube and DoubleClick families. Cookies not matching the list, including first-party session and language cookies, were counted in the total but not as tracking.
- *Consent platform and banner.* A CMP was recorded when a signature of one of thirteen named platforms matched the `src` of a script, iframe or link element, when one of a fixed white-list of CMP global objects was defined, or when `window.__tcfapi` was a function. A banner was recorded when a fixed or sticky element of non-zero height below 600 px contained text matching cookie, consent or privacy. Both are heuristics and their false-negative rate was not measured.
- *Accessibility.* axe-core 4.13 [14] was injected into the loaded page and run over the whole document with `resultTypes` set to violations, passes and incomplete. The rule set was selected by tag: `wcag2a`, `wcag2aa`, `wcag21a`, `wcag21aa`, `wcag22a` and `wcag22aa`, plus `wcag2aaa` and `wcag21aaa`. The reference standard is therefore WCAG 2.2 at levels A and AA [67] together with the level-AAA rules of WCAG 2.0 and 2.1 [65, 66]; no WCAG 2.2 level-AAA rules were enabled, because axe-core 4.13 does not implement them.
- *Derived measures.* Each violation was assigned its success criterion, its principle and its conformance level from the axe tags, with the highest level taken when a rule carries several. A site was recorded as *reaching level A* when it produced no level-A violation, and as having *no automatic failure* when it produced no violation at any enabled level. The number of checks axe flags for manual review was recorded per site but is not analysed here.

Two properties of this protocol matter for interpretation and are not incidental. First, enabling the level-AAA rules is a substantive choice: the dominant AAA rule is enhanced contrast (success criterion 1.4.6, requiring a 7:1 ratio), and any statement about the AAA share of failures is conditional on that rule being active. Section 4.4 therefore reports the level distribution both with and without the AAA rules. Second, the single vantage point in Ecuador is what the pre-consent measurement means: what is observed is what an Ecuadorian visitor experiences, not what a visitor located in the European Union would experience.

3.6 Multi-vantage replication

Because the audit described above observed from one location only, it could not distinguish two possibilities with opposite legal readings: that sites treat every visitor alike, or that they present different interfaces depending on where the visitor is. To separate them, the live audit was replicated from four vantage points on 15 August 2026, with every other parameter unchanged.

The Ecuadorian vantage point was a residential connection in Cuenca (ETAPA EP, autonomous system AS27668) with no virtual private network (VPN). The other three were reached through a commercial VPN: Frankfurt, Germany (DigitalOcean AS14061 for two passes, Limestone Networks AS46475 for one); London, United Kingdom (Zenlayer AS21859); and Miami, United States (Limestone Networks AS46475). Three passes were run per vantage point, twelve in total, and each site's outcome was consolidated across passes by majority rule. The public Internet Protocol (IP) address and its geolocation were verified at the start of every pass, after every 25 sites and at the end, giving seven checks per pass; all 84 checks matched the declared vantage point, and the recordings are released with the data. One further pass was aborted automatically by that mechanism when the VPN had not yet switched country, and was discarded.

Two design properties of this replication matter. First, the Ecuadorian and United States vantage points are both outside the European Economic Area but differ in network type, residential against data centre, whereas the German, British and United States endpoints are all data-centre addresses; the German-versus-United States contrast is therefore free of any residential-versus-data-centre artefact. Second, three quantities that cannot depend on consent logic were used as controls: the number of WCAG violations, the number of failing nodes and the volume of cookies that are not attributable to any advertising vendor.

The tracking-cookie taxonomy was extended for this analysis after inspection of the cookie names observed. The original list covered Google, Meta, TikTok, Microsoft Clarity, Hotjar, Matomo, Baidu and several smaller families; it did not cover the LinkedIn Insight Tag, the full Microsoft advertising family, TapAd, StackAdapt, Snapchat or Sourcebuster. The extended list, given in full in Appendix A, is applied by reclassifying the stored cookie names, so no re-measurement was required and every reported figure derives from the same recorded observations. The extended list is used throughout this article, including in the group comparison of Section 4.2, so that one taxonomy governs every reported figure. Its effect on that comparison was checked rather than assumed, and it is small but not null: reclassification moves one Ecuadorian site across the tracking threshold, because its only matching cookies belong to the Sourcebuster family added at this point, and leaves every other site in both groups unchanged. The Ecuadorian count therefore rises from 40 to 41 of 63 while the benchmark count stays at 43 of 63, and Table 1, Figure 2 and the intervals, tests and equivalence bounds of Section 4.2 are computed on that basis. The extension matters considerably more at the level of individual cookies, which is what Section 4.6 attributes to vendors.

3.7 Statistical analysis

Proportions are reported with 95% Wilson score intervals. Group differences are reported as risk differences with 95% Newcombe intervals and as odds ratios with 95% Woolf intervals, using the Haldane–Anscombe correction of 0.5 where a cell is empty. Significance is assessed with the two-sided Fisher exact test, which is appropriate throughout given the expected cell counts in the regional breakdown. Because twelve indicators are compared, p values are adjusted within that family by the Holm–Bonferroni procedure and both raw and adjusted values are reported.

For the indicator on which no difference was found, an equivalence test was performed rather than an argument from a non-significant p value. Two one-sided tests (TOST) were run at margins of 10, 15 and 20 percentage points. The margins were not pre-specified, which is stated openly, and the result is reported as a bound on what the design can establish rather than as a demonstration of equivalence.

Vantage points are compared with the exact McNemar test, which is the appropriate test when the same site is observed under two conditions, and with Cochran's Q for the four-way comparison; discordant-pair counts and exact p values are reported alongside every such comparison. Multiplicity is handled in two separate families, each declared before the analysis it governs: the twelve indicators of the group comparison, and the six pairwise comparisons between vantage points. The Holm–Bonferroni procedure is applied within each family, the adjusted value governs every claim of difference made in this article, and no claim is made across families. The association between accessibility and pre-consent tracking is analysed within each group separately as well as pooled, because pooling two populations with different marginal distributions on both variables invites Simpson's paradox. Percentages of failing nodes by conformance level are rounded to the nearest integer, which is why the benchmark distribution sums to 101 rather than 100. All analyses were carried out in Python 3 with SciPy; the analysis script is released with the data.

3.8 Ethical considerations

The study involves no human participants and no personal data: the observations are properties of public web pages. It nonetheless carries reputational risk for the institutions observed, and three measures were adopted in response. First, no institution is named in the body of the article as failing an indicator; institution-level results appear only in the appendices, where the coding date, the definition and the limitations of the measure are stated alongside them. Second, the cells of the six institutions whose individual coding carried most weight in the analysis were manually re-verified, and the two corrections found are reported in Section 3.4. Third, the results are framed as a description of observed conduct on one date, not as a determination of legal non-compliance, since, as Appendix B sets out, most of the jurisdictions studied impose no general duty of prior consent for cookies.

A fourth consideration applies to the commercial third parties named in Section 4.6. They are named because their observed conduct is the finding of that section, not as an allegation: what is reported is the cookie state recorded on public pages on a single date, the number of sites is given alongside every name, and no inference is drawn about any vendor's internal configuration, intent or legal position. The observations, the classification rules and the per-site output are released with the data, so that any vendor named can reproduce or contest the measurement.

The script observed a conservative load profile, one request sequence per site with no crawling beyond the home page, and did not attempt to circumvent access controls.

4 Results

Table 1 reports every indicator for both groups with the risk difference, the odds ratio and the Holm-adjusted p value; the Wilson intervals for each proportion are given in the text and displayed in Figure 1. Figure 1 displays the same comparison graphically, grouped by domain. Appendices C and D give the institution-level matrices.

4.1 Privacy governance (RQ1)

The governance differences were the largest and the most robust in the study. A privacy notice was reachable from the home page at 59 of the 63 benchmark sites (93.7%) and at 39 of the 63 Ecuadorian sites (61.9%), a difference of 31.7 percentage points (odds ratio 9.08, adjusted $p < 0.001$). The requirement that the notice name a specific legal instrument reduced both figures, to 74.6% and 49.2% respectively (odds ratio 3.03, adjusted $p = 0.045$).

The widest gap was in the route by which a data subject can act. A privacy contact or data protection officer was identified at 85.7% of benchmark sites and at 38.1% of Ecuadorian sites, a difference of 47.6 percentage points (odds ratio 9.75, adjusted $p < 0.001$). The enumeration of data-subject rights differed in the same direction, 74.6% compared with 55.6%, but this difference did not survive correction for multiple comparisons (adjusted $p = 0.235$) and is reported here as suggestive only.

The regional breakdown of the benchmark group is given in Table 2. It should be read as descriptive: with subgroups of three to five institutions the Wilson interval around a proportion spans more than 50 percentage points, so no regional ranking can be supported by these data. Two patterns are nonetheless worth recording because they are large relative to that uncertainty. The Chinese institutions in the benchmark group are the least likely to publish a first-level notice, two of six, and none published an accessibility

Table 1 Indicator-by-indicator comparison of the benchmark group and the Ecuadorian census ($n = 63$ per group, August 2026). RD: risk difference in percentage points, benchmark minus Ecuador, with 95% Newcombe interval. OR: odds ratio with 95% Woolf interval. p: two-sided Fisher exact test; p_{H}: Holm-adjusted over the twelve comparisons.

Indicator	Benchmark	Ecuador	RD [95% CI]	OR [95% CI]	p_{H}
Privacy governance (documentary)					
Privacy notice published	59/63 (93.7)	39/63 (61.9)	+31.7 [+17.6, +44.7]	9.1 [2.9, 28.2]	<0.001
Framework cited	47/63 (74.6)	31/63 (49.2)	+25.4 [+8.4, +40.4]	3.0 [1.4, 6.4]	0.045
Rights listed	47/63 (74.6)	35/63 (55.6)	+19.0 [+2.4, +34.3]	2.4 [1.1, 5.0]	0.235
Privacy contact or DPO	54/63 (85.7)	24/63 (38.1)	+47.6 [+31.3, +60.4]	9.8 [4.1, 23.3]	<0.001
Pre-consent behaviour (live audit)					
Tracking cookies set	43/63 (68.3)	41/63 (65.1)	+3.2 [−13.0, +19.1]	1.2 [0.5, 2.4]	1.000
At least one cookie set	56/63 (88.9)	55/63 (87.3)	+1.6 [−10.2, +13.4]	1.2 [0.4, 3.4]	1.000
Consent banner shown	22/63 (34.9)	15/63 (23.8)	+11.1 [−4.8, +26.3]	1.7 [0.8, 3.7]	0.961
CMP detected	12/63 (19.0)	14/63 (22.2)	−3.2 [−17.2, +11.0]	0.8 [0.4, 2.0]	1.000
Accessibility					
Accessibility statement	47/63 (74.6)	7/63 (11.1)	+63.5 [+47.9, +74.2]	23.5 [8.9, 61.9]	<0.001
No level-A failure	25/63 (39.7)	5/63 (7.9)	+31.7 [+17.2, +44.9]	7.6 [2.7, 21.7]	<0.001
No failure at any level	6/63 (9.5)	0/63 (0.0)	+9.5 [+1.8, +19.3]	14.4 [0.8, 260]	0.193
Transport security					
Valid TLS certificate	63/63 (100.0)	58/63 (92.1)	+7.9 [+0.6, +17.3]	11.9 [0.7, 221]	0.288

Note: percentages in parentheses. “No level-A failure” denotes the absence of automatically detectable level-A violations and is not a claim of WCAG conformance, which automated tools cannot establish [25]. Two cells of the benchmark group were corrected after manual re-verification, as described in Section 3.4. The tracking indicator is computed under the extended cookie taxonomy of Appendix A.

statement; the same absence of an accessibility statement holds for the five institutions grouped as Rest of Asia, so it is not a feature of the Chinese subgroup alone. Continental Europe cites a legal framework at nine of ten institutions, but the ten institutions are not homogeneous in the applicable instrument: two are Swiss and are subject to the revised Federal Act on Data Protection [58] rather than to the GDPR. Similarly, the three Canadian institutions are public bodies governed by provincial public-sector statutes [34, 35, 44], not by the federal Personal Information Protection and Electronic Documents Act (PIPEDA) [49], which applies to commercial activity, so a single Canadian regime cannot be assumed.

One further documentary indicator is recorded in Appendix C and is reported here rather than compared. A dedicated cookie policy, which is a published document and therefore distinct from the live banner measurement of Section 4.2, was found at 14 of the 63 benchmark institutions (22.2%). It is concentrated in the United Kingdom, at 5 of 7 institutions, and in continental Europe, at 4 of 10, and absent from all six Chinese institutions and from all five in the Rest of Asia group. It is excluded from Table 1 because the Ecuadorian matrix records no counterpart, so the indicator has no comparison group; it is retained in the appendix because it is the documentary complement of the live measurement and readers may wish to relate the two.

4.2 Pre-consent behaviour (RQ2)

Figure 2 reports the live measurements. Observed from Ecuador, the two groups behaved similarly. At least one cookie was set before any consent interaction at 88.9% of benchmark sites and 87.3% of Ecuadorian sites; tracking cookies specifically were set at 68.3% and 65.1%. The difference on the tracking indicator was 3.2 percentage points in favour of Ecuador, with a 95% interval of $[-13.0, +19.1]$ percentage points and an adjusted p of 1.000. A visible consent banner was, in both groups, a minority phenomenon (34.9% and 23.8%), and a consent-management platform was detected slightly more often in Ecuador (22.2%) than in the benchmark group (19.0%), a difference well within sampling variation.

All figures in this section are computed under the extended cookie taxonomy of Appendix A,

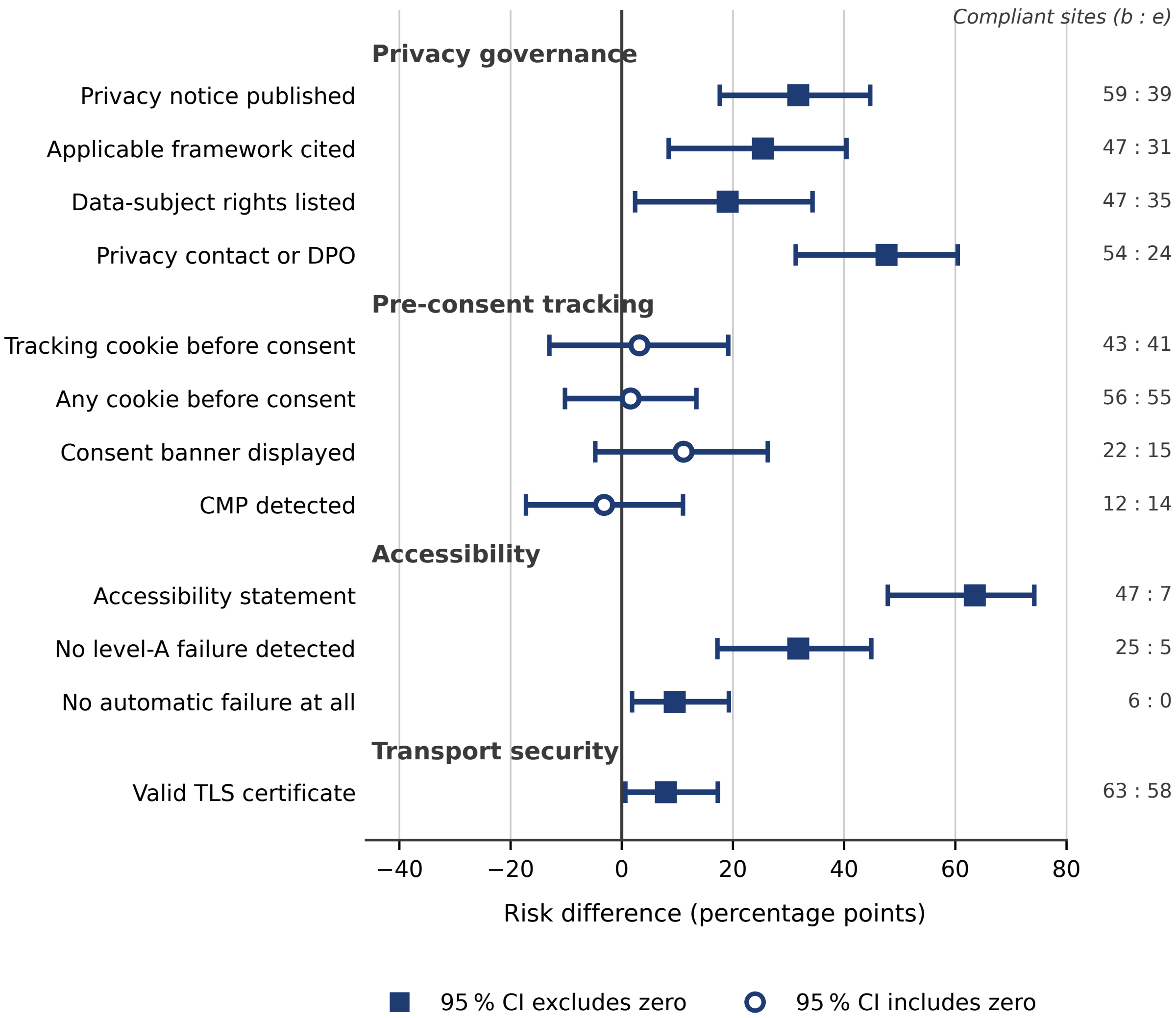

Fig. 1 Risk differences between the benchmark group and the Ecuadorian census, with 95% Newcombe intervals, grouped by domain. Filled squares mark intervals that exclude zero; open circles mark intervals that include it. Counts of compliant sites out of 63 are given at the right. Positive values indicate a higher proportion in the benchmark group.

Table 2 Benchmark group by region: institutions meeting each documentary indicator out of the regional total. Percentages and intervals are omitted deliberately: with n between 3 and 24 they would suggest a precision the data do not have.

Region	n	Notice	Framework	Contact/DPO	Accessibility
United States	24	24	17	23	22
United Kingdom	7	7	4	5	7
Continental Europe	10	10	9	10	8
China	6	2	2	3	0
Hong Kong SAR	3	3	3	2	3
Rest of Asia	5	5	4	4	0
Australia	5	5	5	4	4
Canada	3	3	3	3	3
Total	**63**	**59**	**47**	**54**	**47**

as is the whole of this article. Under the original list the Ecuadorian tracking count would be 40 of 63 rather than 41, a difference of one site, and the benchmark count would be unchanged; no conclusion in this section depends on that choice.

The absence of a detectable difference must not be read as a demonstration of equivalence. With 63 sites per group and proportions near 0.65, the smallest difference this design could detect with 80% power at $\alpha = 0.05$ is approximately 23 percentage points; detecting a true difference of the size observed here would require roughly 3400 sites per group. Two one-sided tests confirm the point directly: equivalence is rejected at a margin of ± 10 percentage points ($p = 0.208$) and at ± 15 ($p = 0.079$), and is only supported at a margin of ± 20 ($p = 0.023$), which is too wide to be substantively interesting. Demonstrating equivalence within ± 10 points would require approximately 350 sites per group. The correct statement of this result is therefore that *the study found no evidence of a difference in pre-consent tracking, and had no power to establish that the groups are equivalent.*

A further and more important qualification concerns what a null result here can mean at all. Because all observations were taken from Ecuador, the benchmark figure conflates two possibilities that carry opposite legal readings: that European institutions do not condition their consent interfaces on the visitor's location and simply track everyone, or that they do condition them and therefore show a European visitor a different interface from the one recorded here. The present design cannot distinguish the two, and Section 5 sets out why the distinction matters and how it should be resolved.

4.3 Declared accessibility and transport security (RQ3, part one)

An accessibility statement or a set of accessibility tools was published by 47 of the 63 benchmark institutions (74.6%) and by 7 of the 63 Ecuadorian institutions (11.1%), the largest single difference in the study (odds ratio 23.50, adjusted $p < 0.001$). Concrete statutory duties plausibly account for part of this gap: 22 of the 24 United States institutions publish a statement, in a jurisdiction with the ADA and Section 508 [62, 63], and 8 of the 10 continental European institutions do so under the European Accessibility Act [22]. Ecuador's low figure is consistent with the regional lag documented over the past decade [2, 9].

Transport security was near-universal in both groups: a valid TLS certificate was present at all 63 benchmark sites and at 58 of the 63 Ecuadorian sites (92.1%; adjusted $p = 0.288$). We stress that this indicator measures confidentiality in transit only. A site with an impeccable certificate chain may transmit visitor data to any number of third parties, and five of the twelve indicators in Table 1 show precisely that pattern.

The transparency section required of Ecuadorian public bodies by the Organic Law on Transparency and Access to Public Information (Ley Orgánica de Transparencia y Acceso a la Información Pública, LOTAIP) was present at 47 of the 63 Ecuadorian institutions (74.6%). It is reported here for completeness because it is recorded in Appendix D; it is not used in any comparison, since it has no counterpart in the benchmark group.

4.4 Measured WCAG conformance (RQ3, part two)

Table 3 reports the audit results and Figure 3 the distribution of failing nodes by conformance level.

Neither group performed well. No Ecuadorian site and only 9.5% of benchmark sites produced no automatic failure at any enabled level, and the median number of distinct rules violated was four in Ecuador compared with two in the benchmark group. The distinction that matters, however, is where the failures fall. In the benchmark group 19% of failing nodes were at level A and 60% at level AAA, the latter dominated by enhanced contrast; in Ecuador 45% were at level A. Excluding the AAA rules entirely and re-normalising over levels A and AA preserves the ordering, 46.0% compared with 61.1% at level A, which indicates that the qualitative finding is not an artefact of having enabled a single demanding rule.

Two readings that the aggregate figures invite are not supported by these data, and both are worth ruling out explicitly because a benchmarking design encourages them. It is *not* the case that no institution is free of automatic failures: six benchmark institutions were. And it is *not* the case that the benchmark group fails only at the

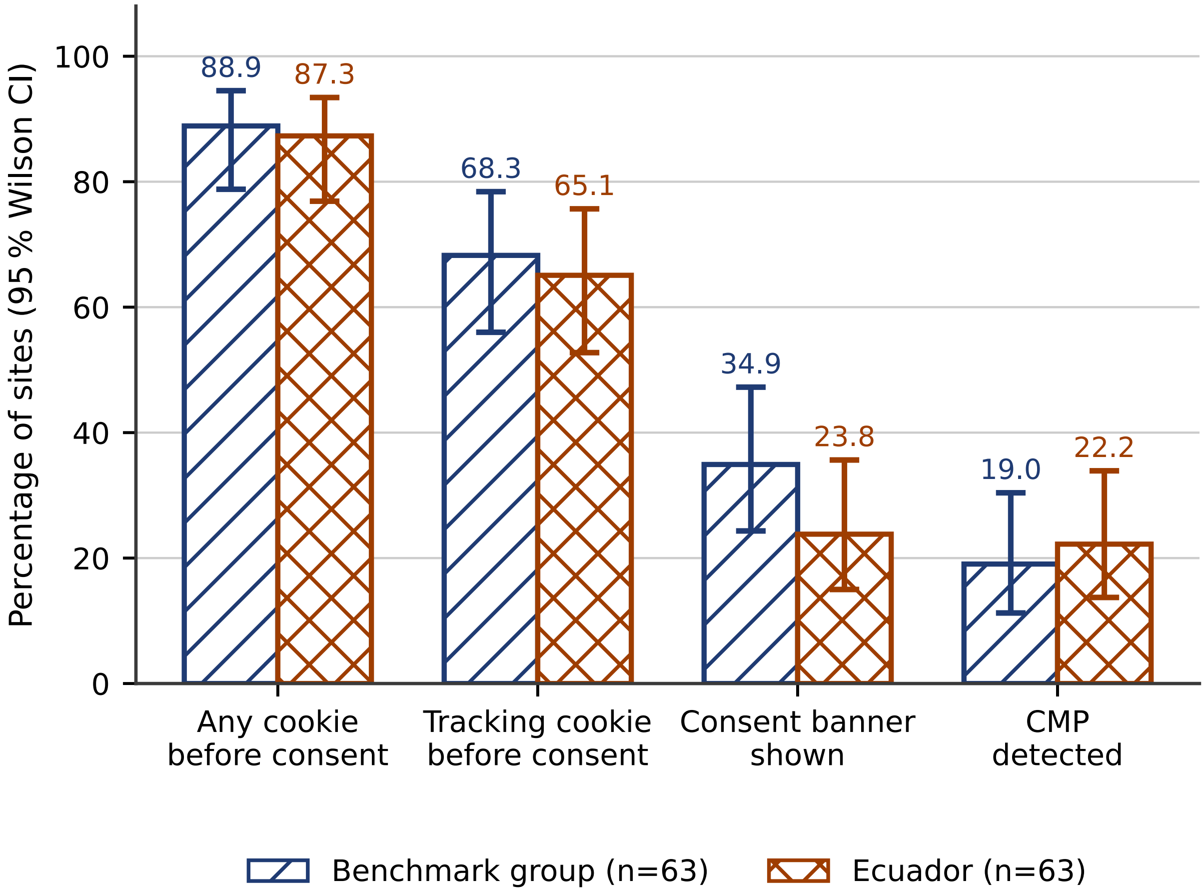

Fig. 2 Cookie state recorded before any interaction with a consent interface, with 95% Wilson intervals. All measurements were taken from a single vantage point in Ecuador in August 2026; a European visitor would not necessarily observe the same behaviour on the European sites of the benchmark group.

level of excellence: 60.3% of benchmark sites produced at least one level-A failure, so the majority of the reference group does not clear the basic tier either. What the data support is narrower and still worth reporting: the two groups differ in the *distribution* of their failures, not in the presence of failures.

The single clearest instance is success criterion 2.4.4, which requires that every link have text a screen reader can announce; 81.0% of Ecuadorian sites and 23.8% of benchmark sites violated it. Insufficient contrast (1.4.3) followed the same pattern, 79.4% compared with 25.4%. The most affected principle in both groups is perceivability, through contrast; Ecuador adds an operability deficit through links and controls that is marginal in the benchmark group. These are the same dominant error types reported for other public-sector portals [4, 30].

Two cautions apply to the level-based measures. First, the distribution in Figure 3(a) is computed over failing *nodes*, which are not independent within a site: a single template with several thousand low-contrast elements can dominate its group's distribution. Second, the distribution is compositional, so a group with more level-A failures must, arithmetically, show a smaller AAA share. Both are reasons to treat the level distribution as descriptive; the site-level measures in the upper half of Table 3 are the inferentially safe ones, and they point in the same direction.

Table 3 Automatically detectable WCAG results (axe-core 4.13, August 2026, home page only). Percentages are of sites unless stated otherwise. The rule set is specified in Section 3.5.

Measure	Benchmark	Ecuador
Median distinct rules violated per site	2	4
Sites with no automatic failure at any level	9.5%	0.0%
Sites with no automatic level-A failure	39.7%	7.9%
Links without an accessible name (2.4.4, level A)	23.8%	81.0%
Insufficient contrast (1.4.3, level AA)	25.4%	79.4%
Images without a text alternative (rule `image-alt`, success criterion 1.1.1, level A)	15.9%	30.2%
Distribution of failing nodes by level, all rules enabled		
Level A	19%	45%
Level AA	22%	29%
Level AAA	60%	26%
Same distribution with AAA rules excluded and re-normalised		
Level A	46.0%	61.1%
Level AA	54.0%	38.9%

Note: the median is of distinct rules violated, not of failing nodes. Level percentages are of each group's total failing nodes and are rounded to the nearest integer, which is why the benchmark distribution sums to 101%.

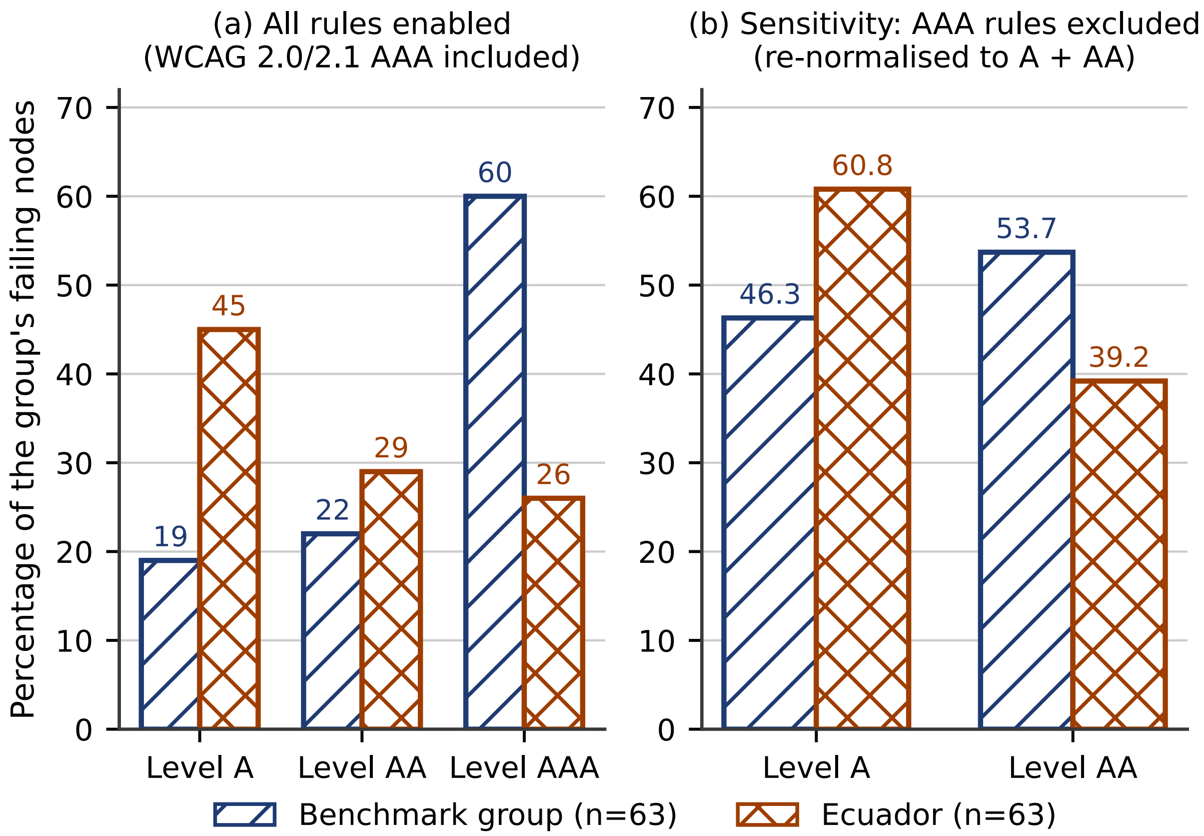

Fig. 3 Distribution of failing nodes by WCAG conformance level. Panel (a) uses the full rule set of this study, which includes the level-AAA rules of WCAG 2.0 and 2.1. Panel (b) excludes those rules and re-normalises over levels A and AA, showing that the ordering of the two groups at level A is preserved when the enhanced-contrast rule is not counted.

4.5 Do the two commitments co-occur? (RQ4)

Across all 126 sites, the Spearman correlation between the number of failing accessibility nodes and the number of pre-consent tracking cookies is 0.05. That coefficient has a 95% confidence interval of $[-0.13, +0.22]$ and $p = 0.578$, and the design could only have detected a correlation of about 0.25 with 80% power. The pooled coefficient is therefore compatible with no association and with a weak positive association alike, and it cannot support a claim of independence.

The pooled coefficient is also the wrong estimand, because it mixes two populations whose marginal distributions differ on both variables. Table 4 gives the within-group cross-classification of the two dichotomised measures. Within the benchmark group the association is, if anything, slightly negative (odds ratio 0.54, Fisher $p = 0.408$); within the Ecuadorian group it is essentially absent (odds ratio 1.27, Fisher $p = 1.000$); the Mantel–Haenszel estimate adjusted for group is 0.67. None of these is distinguishable from no association at this sample size.

The defensible conclusion from RQ4 is a negative one, and it is worth stating precisely because the temptation is to overstate it. These data provide no evidence that institutions which invest in accessibility also restrain tracking, and no evidence of the reverse. They are not sufficient to establish that the two are independent; establishing that would require either a substantially larger sample or a continuous, better-powered pair of measures.

4.6 Whose location determines protection? (RQ5)

The four-vantage replication yielded 121 of the 126 sites measured successfully in all three passes at all four vantage points. The five sites that fall out are named here, because a paired design cannot impute them and the reader is entitled to know which they are. Each failed to load on at least one pass, in every case with a connection timeout or with the 45 s navigation limit exceeded, that is, because the server did not respond rather than because the audit was interrupted. They are Université Paris-Saclay, which failed on one British pass; ESPOL, on two German passes; UCSG, on two Ecuadorian passes; UDET, on one German and all three British passes; and UNEMI, on all three Ecuadorian passes and one United States pass. Four of the five are Ecuadorian, which is itself a finding about the availability of those portals and is taken up in Section 5.5. The 121 sites retained comprise 62 of the benchmark group and 59 of the Ecuadorian census. The three control quantities behaved as required: the mean number of WCAG violations per site was 3.53, 3.51, 3.54 and 3.50 at the Ecuadorian, German, British and United States vantage points respectively, a spread of 1.1% between the extremes, confirming that the pages themselves rendered identically. Between-pass agreement on the tracking indicator, computed pairwise within each vantage point over the 121 sites, ranged from 95.9% to 100%, so three passes were sufficient.

Pre-consent tracking, by contrast, differed by visitor location. It was observed at 82 of the 121 sites from Ecuador (67.8%, 95% CI 59.0–75.4), at 79 from the United States (65.3%), at 76 from the United Kingdom (62.8%) and at 73 from Germany (60.3%, 51.4–68.6). Cochran's Q over the four vantage points gives $Q = 15.88$, 3 df, $p = 0.0012$.

The paired comparisons in Table 5 locate that difference, and the multiplicity policy of Section 3.7 is applied to the six of them as a family. Ecuador differs from Germany, with 9 discordant pairs all in the same direction (exact $p = 0.0039$; Holm-adjusted $p = 0.023$), and this contrast is the finding of the section: it is also the contrast that Section 3.6 identified as free of any residential-versus-data-centre artefact. The difference between Ecuador and the United Kingdom does not survive correction within the family (raw $p = 0.031$, adjusted $p = 0.156$) and is reported as suggestive only. No other pair is distinguishable: the smallest adjusted value among the remaining four is 0.281, for the United States against Germany.

The conclusion does not depend on that inclusion rule. Requiring success in all three passes at every vantage point is the strictest of the three defensible criteria, and the analysis was repeated under the two weaker ones. Requiring at least two valid passes per vantage point admits Paris-Saclay and gives $n = 122$, with every discordant-pair count, every exact p value and Cochran's Q identical to those in Table 5. Requiring at least one valid pass admits UCSG and ESPOL as well and

Table 4 Cross-classification of automatically detectable level-A conformance against pre-consent tracking, within each group. OR: odds ratio of conformance given restraint in tracking, with 95% interval.

Group	A^+T^-	A^+T^+	A^-T^-	A^-T^+	**OR [95% CI]**	p
Benchmark	6	19	14	24	0.54 [0.17, 1.68]	0.408
Ecuador	2	3	20	38	1.27 [0.20, 8.21]	1.000

Note: A^+ no automatically detectable level-A failure, A^- at least one; T^- no tracking cookie before consent, T^+ at least one. p from the two-sided Fisher exact test. In Ecuador, 2 institutions of 63 met both criteria. Under independence, $63 \times 0.079 \times 0.349 \approx 1.7$ would be expected, so this figure is exactly what chance predicts and carries no information about coupling.

gives $n = 124$, under which the marginals become 84, 80, 78 and 74 sites, Cochran's Q rises to 16.42 ($p = 0.00093$) and the Ecuador–Germany contrast strengthens to ten discordant pairs, all in the same direction (exact $p = 0.0020$; Holm-adjusted $p = 0.012$). UNEMI and UDET cannot be admitted under any rule, since neither has a single valid observation at one of the four vantage points. The ordering of the four vantage points and the significance of the Ecuador–Germany contrast are therefore invariant to this choice, and the strictest rule is the one reported because it is the most conservative.

Three tracking figures for Ecuador appear in this article and are not interchangeable, so the relation between them is stated here rather than left to the reader. Table 1 reports 41 of the 63 Ecuadorian sites and 43 of the 63 benchmark sites, that is 84 of the 126 in all (66.7%), from a single pass on 14 August 2026. The present section reports 82 of the 121 sites observed successfully at every vantage point (67.8%), consolidated over three passes on 15 August 2026. Both use the extended taxonomy, and the two figures differ in site set, consolidation rule and date, and neither supersedes the other: the first supports the between-group comparison, the second the within-site comparison across vantage points.

The mechanism becomes visible once the cookies are attributed to the vendor that sets them. Figure 4 shows the share of sites at which each vendor set at least one cookie before any consent interaction. On 15 August 2026 three vendors set no cookies at all at the German and British vantage points while setting them at the Ecuadorian and United States ones, on the same pages and within the same pass: Microsoft, whose Clarity, advertising and Universal Event Tracking (UET) cookies appeared at 14 sites from Ecuador and 14 from the United States and at none from Germany or the United Kingdom; TapAd, at 4 and 4 sites and at none; and StackAdapt, at 2 and 2 sites and at none, a count too small to support anything beyond its own description. The LinkedIn Insight Tag followed the same pattern in attenuated form, at 14 and 14 sites compared with 3 and 7. Google Analytics, Meta and Adobe reduced their footprint only moderately and did so similarly at all four vantage points. These are observations of conduct on one date; they are not statements about what any vendor decided or intended.

Two features of this pattern deserve emphasis because they are not what the institutional framing would predict. First, the suppression is not confined to European institutions: Ecuadorian, Chinese and United States universities that embed these tags also stop setting the corresponding cookies when the visitor is in Germany or the United Kingdom. The behaviour therefore belongs to the vendor, not to the institution or to the law that binds it. Second, the United Kingdom is treated by most vendors as inside the protected perimeter although it is no longer in the European Economic Area, which is consistent with those vendors operating a country list rather than a legal test. TikTok and Baidu are the exceptions: both set materially more cookies for British visitors than for German ones, 10 sites compared with 8, and 2 compared with 0, respectively, so the perimeter is vendor-specific rather than common.

One caveat applies. Cookies not attributable to any advertising vendor also fell at the European and British vantage points, by 25.5% and 21.2% relative to Ecuador, whereas the United States vantage point differed by only 0.8%. That residual is smaller than the 43.9% and 37.7% fall in vendor cookies and it does not affect the rendering controls, but part of it is likely to be the Cloudflare bot-management cookie and the cookies set by consent platforms themselves, whose behaviour also varies with the visitor's region. The residual is reported here rather than adjusted away.

Table 5 Exact McNemar tests for pre-consent tracking between vantage points ($n = 121$ sites observed at all four). a: sites tracking at both vantage points; b: sites tracking at the first but not the second; c: the converse; d: sites tracking at neither. Only discordant pairs carry information. p_{H}: Holm-adjusted over the six comparisons of this family.

Comparison	a	b	c	d	Discordant	p	p_{H}
Ecuador vs Germany	73	9	0	39	9	0.0039	0.023
Ecuador vs United Kingdom	76	6	0	39	6	0.031	0.156
United States vs Germany	72	7	1	41	8	0.070	0.281
United Kingdom vs Germany	73	3	0	45	3	0.250	0.750
Ecuador vs United States	79	3	0	39	3	0.250	0.750
United States vs United Kingdom	75	4	1	41	5	0.375	0.750

Note: rows are ordered by raw p. The marginal totals implied by the table are 82 sites tracking from Ecuador, 79 from the United States, 76 from the United Kingdom and 73 from Germany, which are the counts reported in the text.

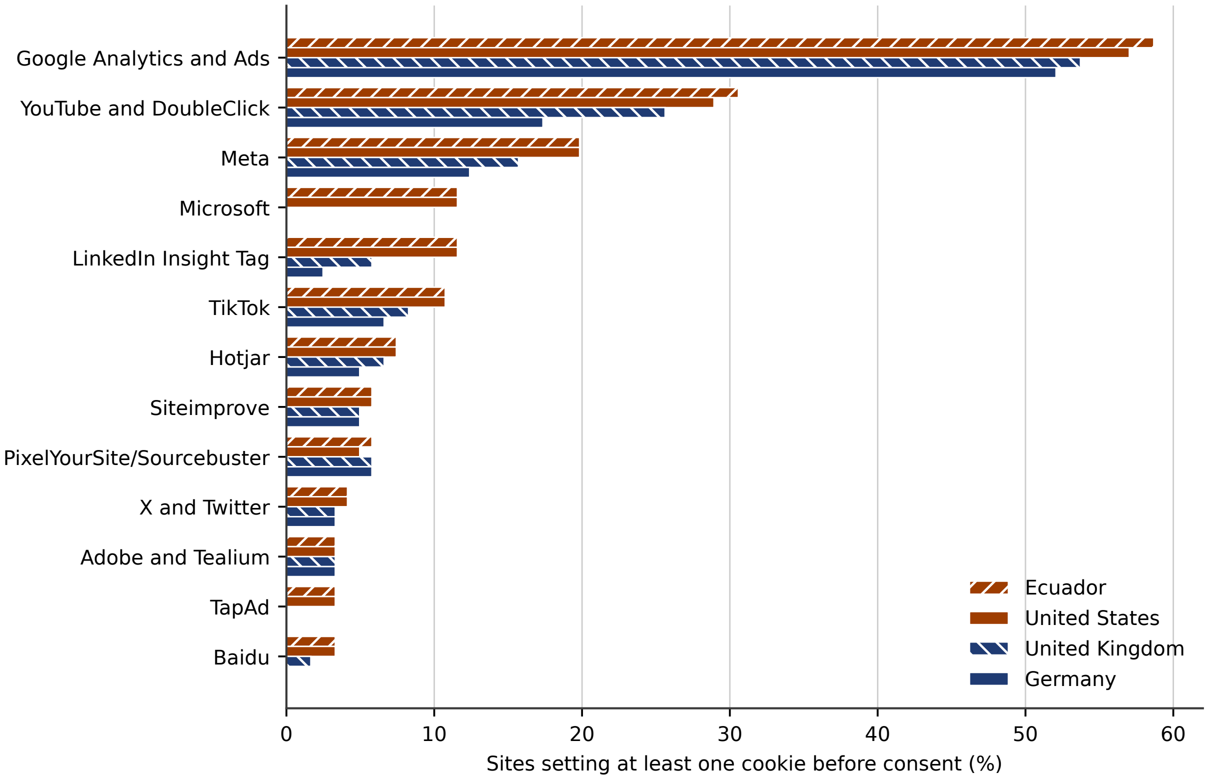

Fig. 4 Share of sites at which each advertising vendor set at least one cookie before any consent interaction, by vantage point ($n = 121$ sites, majority of three passes, August 2026). Ochre marks the two vantage points outside the European Economic Area and navy the two inside the perimeter that vendors apply. Vendors are ordered by their prevalence as seen from Ecuador; the five present at fewer than four sites from Ecuador are omitted, namely StackAdapt, Snapchat, ContentSquare, Pinterest and Matomo.

5 Discussion

5.1 Interpretation

Three findings survive the analysis of Section 4, and they are of different kinds.

The first is a governance gap that is large, robust to multiple-comparison correction and consistent across four indicators. The Ecuadorian shortfall is not principally the absence of documents but the absence of the elements that make a document actionable: a named instrument, an enumeration of rights and, above all, an identified route through which a person can exercise them. The 47.6-point gap on the privacy contact is the single most consequential number in the study, because a right without a channel is not exercisable. That benchmark institutions perform well here is not evidence of superior ethics; it is the expected consequence of legal regimes that impose the duty explicitly, and it suggests that the same is achievable in Ecuador as the LOPDP and its regulation move into their enforcement phase [5, 53].

The second is an accessibility gap whose interesting property is its shape. Both groups fail; the majority of the benchmark group fails at the basic tier too. What differs is that Ecuadorian failures concentrate at level A, where the remedies are inexpensive and well understood, principally accessible link text and adequate contrast, and where the consequences for users of assistive technology are most severe. This is the most immediately actionable finding of the study and the one that requires no new legislation to act on.

The third is a null result on pre-consent tracking that must be reported as such. Observed from Ecuador, roughly two thirds of the institutions in both groups set tracking cookies before any consent interaction. The study found no difference; it also had no power to establish equivalence, and Section 4.2 states the bound explicitly. The finding that survives is therefore not "there is no gap" but "the gap, if any, is not large, and it is certainly not in the direction that the reputational ordering of the two groups would predict".

5.2 Territorial scope, and who actually enforces it

One explanation of the benchmark group's behaviour suggests itself immediately, and it is wrong twice over: that the protection of the GDPR stops at the Union's border, so that a student consulting a European university's portal from Ecuador is as exposed as on a domestic site. It is wrong in law, and the four-vantage replication shows that it is also wrong in fact, though not in the direction that framing would predict.

The legal point first. Article 3(1) of the GDPR applies to processing in the context of the activities of an establishment of a controller in the Union *regardless of whether the processing takes place in the Union or not*, and the European Data Protection Board (EDPB) has confirmed that the location of the data subject is irrelevant to that limb [18, 21]. A German or French university that sets marketing cookies in the browser of a visitor located in Ecuador is therefore acting within the material and territorial scope of the Regulation. Article 3(2) extends the Regulation to controllers not established in the Union that offer goods or services to data subjects *who are in* the Union, so that limb keys on the data subject's location rather than the controller's. The narrower and defensible point concerns a different instrument: Article 5(3) of the ePrivacy Directive, as transposed nationally, is articulated around the terminal equipment of a user, and the EDPB's guidance on its technical scope is the appropriate reference [19, 20].

The measurement then shows something the institutional framing does not anticipate. Protection does vary with the visitor's location, and substantially, but it is not the universities that vary their behaviour. It is the advertising vendors embedded in their pages. Microsoft, at 14 sites, TapAd, at 4, and StackAdapt, at 2, set no cookies at all at the German and British vantage points and set them normally at the Ecuadorian and United States ones, and they did so on Ecuadorian, Chinese and United States university sites as readily as on European ones. The institution is a bystander to this: it embeds a tag, and the geolocation logic inside the tag determines what the visitor receives.

Three consequences follow, and together they are the most original contribution this study can make.

The first is that compliance is being executed at the wrong layer. A duty that European law places on the controller, the university, is in practice discharged by a processor that the university did not configure and does not audit, on the basis of a geolocation lookup. That may satisfy the letter of the obligation for European visitors while leaving the controller without any knowledge of what its own site does.

The second is that the boundary is commercial, not legal. The United Kingdom left the European Economic Area but most vendors still place it inside the protected perimeter, while TikTok and Baidu do not. A person's protection therefore depends on which country lists a set of private firms happen to maintain, and those lists are neither published, nor harmonised, nor subject to the review that a legal instrument would receive.

The third is distributive, and it is the point that connects this finding to the concern of universal access with which this article began. The population that receives no protection is the one located outside the vendors' perimeter, which includes the entire international applicant pool of Latin America, Africa and much of Asia. An Ecuadorian applicant browsing a German university's admission pages is tracked; a German applicant browsing an Ecuadorian university's pages is not. The asymmetry runs in the direction that compounds existing inequality, and it is invisible to any monitoring exercise conducted from inside the perimeter, which is where regulators, auditors and most researchers sit. This is precisely the exclusion by geopolitical and economic position that Abascal et al [1] argue universal accessibility must address alongside exclusion by ability, and it appears here in an unusually tractable form: the perimeter is a list of countries, so the excluded population can be enumerated and the boundary can be measured.

5.3 Implications for practice

For Ecuadorian institutions, the ordering implied by these results is unusually clear. The first action is to identify a privacy contact by role and address on the first-level notice; it is the cheapest intervention in the list and it closes the largest gap. The second is to name the applicable instrument and enumerate the rights the LOPDP confers, together with the procedure to exercise them, replacing generic references to "applicable legislation". The third is to remediate the level-A accessibility failures, beginning with link text and contrast, which together account for the majority of the level-A nodes observed. The fourth, and the least urgent on the evidence collected here, is the consent interface: adding a banner without configuring it to block before consent changes the measured indicator without changing the user's exposure, a substitution that the measurement literature has documented repeatedly [7, 56] and that the present data cannot rule out in either group.

For institutions in the benchmark group, the finding that 60.3% of them produce at least one level-A failure, and that two thirds set tracking cookies before consent when observed from outside the Union, is the relevant one. Reputational standing is not evidence of either accessibility or privacy performance. A further implication follows from Section 4.6 and applies to every institution in both groups: an administrator who tests their own site from their own office sees only what their own location entitles them to see. Verifying what an international applicant experiences requires testing from that applicant's region, which is neither difficult nor expensive and is, on this evidence, not being done.

5.4 Implications for policy

Two implications follow for regulators. First, for the Ecuadorian data protection authority, the indicators that separate the two groups most sharply are exactly those that are verifiable at low cost from outside the institution, so a periodic supervised self-assessment against a published matrix is administratively feasible; the codebook in Appendix A is offered for that purpose. The accompanying risk is well known and should be anticipated: an indicator that becomes a target is satisfied at the surface, and the cookie banner is the clearest instance of an indicator that can be met without any change in behaviour.

Second, the pattern documented in Section 4.6 concerns European supervisory authorities rather than Ecuadorian ones, and it raises a question that supervision as currently practised does not ask. Where a controller's compliance with Article 5(3) is delivered de facto by a processor's geolocation logic, the controller cannot demonstrate accountability under Article 5(2) for conduct it neither configured nor observes. Supervisory testing conducted from inside the perimeter will find that controller compliant. We suggest that audits of consent behaviour be performed from outside the European Economic Area as a matter of routine, and that the country lists operated by major tag vendors be treated as a supervisable object in their own right.

5.5 Threats to validity

Construct validity. Pre-consent tracking is measured uniformly across eleven jurisdictions whose obligations differ radically; in the United States there is no general prior-consent duty, and Ecuadorian law does not incorporate an equivalent of Article 5(3) of the ePrivacy Directive. Appendix B maps each indicator to the concrete obligation, if any, in each jurisdiction, and the results in Section 4.2 are stated as observed conduct, not as compliance findings. "No level-A failure detected" is a statement about the machine-testable subset and not a conformance claim [25]. Transport security is reported separately from data protection for the reason given in Section 4.3. Cookie-based measurement also misses `localStorage`, `IndexedDB`, fingerprinting, cookieless pixels and server-side tagging, so the tracking figures reported here are lower bounds.

Internal validity. The benchmark group is an elite stratum, not a sample of the world, so resource and jurisdiction effects are fully confounded; no third stratum was measured. Confounders that could be recorded were not, in particular budget, enrolment, site language, content management system and hosting provider, and the last of these is a plausible source of non-independence within the Ecuadorian group, where shared templates are common. The institution type recorded in Appendix D, public, co-financed or self-financed, is not analysed here.

Reliability. Documentary coding was performed without double coding and without an inter-coder reliability coefficient. The verification sub-study of Section 3.4 found two errors in 24 re-examined cells; because those cells were selected purposively, the figure is neither an upper nor a lower bound on the error rate of the full matrix, and it is reported only as evidence that the rate is not negligible. The coding of items requiring judgement across several languages, among them Chinese, Japanese, Korean, German, French and Swedish, is the most exposed part of the design. The asymmetry with the live measurements is itself a limitation: geolocation was verified 84 times and the cookie state consolidated over three passes, whereas every documentary cell rests on a single reading by one coder. Partial codes were treated as non-compliant, a convention that penalises institutions with segmented notices; the percentages were not recomputed under the alternative convention. Codes for non-verifiable sites remained in the denominator, which depresses the Ecuadorian percentages specifically, since all such cases occur in that group.

Vantage-point artefacts. The three non-Ecuadorian vantage points were reached through a commercial VPN and their egress addresses belong to data centres, so vendors that treat such addresses differently could in principle mimic a location effect. Three observations bound this concern: the rendering controls were identical across vantage points to within 1.1%; the German and United States endpoints share the same network type, so their contrast is free of the artefact; and one German endpoint was replaced by a second provider on a different autonomous system with an indistinguishable result. The residual fall in non-vendor cookies at the European and British vantage points, reported in Section 4.6, is not fully explained and is the weakest point of this part of the design. All four vantage points were also measured on a single day, so the replication describes vendor behaviour on that date. A further limitation concerns which sites the replication could retain. Five of the 126 failed to respond on at least one of the twelve passes and are excluded from the paired analysis, and four of those five are Ecuadorian; the loss is therefore not random with respect to group, and it points in the direction of lower availability among the Ecuadorian portals. Because the sensitivity analysis reported in Section 4.6 leaves the conclusion unchanged, the effect on the vantage-point comparison is negligible, but the differential availability itself is worth recording as an observation about the infrastructure of the two groups.

Measurement stability. The documentary review remains a single observation. Pre-consent cookie state is not deterministic: it depends on A/B tests, ad auctions, content delivery network (CDN) node and deployment timing. For the live audit this was addressed by the three passes per vantage point, whose between-pass agreement ranged from 94.4% to 100%; for the group comparison of Section 4.2, which rests on the August single-pass measurement, it was not. Only the home page was audited, whereas WCAG conformance is defined over pages and complete processes, and the transactional paths this article invokes as its motivation, admission and enrolment, were not measured at all. The banner and CMP detectors are heuristics whose false-negative rates are unknown.

Statistical conclusion validity. Twelve comparisons were corrected by the Holm procedure and only the adjusted values are used to support claims; the regional breakdown is reported without inference for the reason given in Section 4.1. The equivalence margins in the TOST analysis were not pre-specified. The node-level accessibility distribution is subject to pseudo-replication and compositional constraint, as stated in Section 4.4. Failure counts were not normalised by page complexity, so a large portal will accumulate more failing nodes than a small one without necessarily being less accessible.

External validity. All observations were taken on one date from one country, and the results describe the state of 126 home pages in August 2026. Web content changes; no claim in this article should be read as a durable property of any institution.

6 Conclusions and future work

Measured against a benchmark of 63 top-ranked universities, the 63 Ecuadorian HEIs showed their widest and most robust gaps in privacy governance, above all in the identification of a privacy contact (38.1% compared with 85.7%) and in the publication of a first-level privacy notice (61.9% compared with 93.7%). In accessibility, both groups failed, but the failures were distributed differently: 45% of Ecuadorian failing nodes were at the basic level A compared with 19% in the benchmark group, an ordering preserved when the level-AAA rules are excluded. In pre-consent tracking, observed from Ecuador, the study found no difference between the groups and had no power to establish that they are equivalent. Within each group, measured accessibility and observed restraint in tracking were not associated.

Replicating the live audit from four vantage points then established that pre-consent exposure varies with the visitor's location rather than the institution's. On 15 August 2026 the same 121 pages set tracking cookies for 67.8% of visits from Ecuador and 60.3% of visits from Germany (Cochran Q, $p = 0.0012$; Ecuador–Germany Holm-adjusted $p = 0.023$ over the six pairwise comparisons), and the suppression was executed by advertising vendors: Microsoft, at 14 sites, TapAd, at 4, and StackAdapt, at 2, set nothing at the German and British vantage points while setting cookies at the Ecuadorian and United States ones on the same pages, including on Ecuadorian and Chinese university sites.

These findings support a practical conclusion, a legal one and a distributive one. The practical conclusion is that Ecuador's distinctive shortfalls are in governance and in basic-level accessibility, and both are remediable without new legislation. The legal conclusion is that a duty European law places on the controller is in practice being discharged by processors on the basis of unpublished country lists, which satisfies the visitor in Frankfurt while leaving the controller unable to demonstrate accountability for what its own site does. The distributive conclusion is the one that returns this article to its starting point: the applicants who receive no protection are those outside the perimeter those lists describe, which is to say most of the world's international applicant pool, and their exposure is invisible to anyone testing from inside it.

Four further items follow from the threats stated in Section 5.5: independent double coding of at least 30% of the sample with a Cohen's κ per indicator and full reconciliation of discrepancies; extension of the audit to at least five pages per site, including an admission or enrolment path, with the page as the unit and the site as a random effect; more robust tracking measures than the cookie, in particular the count of distinct third-party domains contacted before consent; and an internal analysis of the Ecuadorian group by institution type, size and platform, for which the necessary variable is already recorded in Appendix D and which is likely to be of greater local value than any international comparison. To these the present results add a fifth: mapping the country lists operated by the major tag vendors, by measuring from one vantage point per candidate jurisdiction, so that the perimeter within which protection is delivered can be described empirically rather than assumed to coincide with any legal boundary.

Supplementary information. The audit script, its configuration, the raw per-site JavaScript Object Notation (JSON) output of the twelve multi-vantage passes together with the location-verification records, the complete cookie inventory with its classification under both taxonomies, the documentary matrix, the codebook and the analysis notebooks are deposited in a public repository under the identifier supplied with this submission. Every figure reported in Section 4.6, including the vendor-level attribution, is reproducible from those files with the analysis script released alongside them.

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Statements and Declarations

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Competing interests. The corresponding author and the third author are affiliated with Universidad Técnica Estatal de Quevedo, which is one of the 63 Ecuadorian institutions assessed in this study and appears in Appendix D. The second author is affiliated with the Ministry of Education of Ecuador, a public body of the same national education system. Neither institution had any role in the design of the study, in the collection or coding of data, in the analysis or in the decision to publish, and the coding of the authors' own institution follows the same codebook and is reported in Appendix D without exception.

Author contributions. G.G.-U.: conceptualisation, methodology, software, formal analysis, writing — original draft. A.Z.P.: investigation, data curation, writing — review and editing. E.D.-M.: investigation, validation, writing — review and editing. All authors read and approved the final manuscript.

Data availability. The datasets generated and analysed during the current study are available in the repository cited under Supplementary information.

Code availability. The audit script and the analysis code are available in the same repository.

Ethics approval and consent to participate. Not applicable. The study analysed publicly accessible web pages and involved no human participants, no animals and no personal data.

Use of generative artificial intelligence. Generative artificial intelligence tools were used to assist with language editing and with the drafting of auxiliary analysis scripts. They were not used to generate, select or interpret data, nor to produce any of the results, tables or figures reported here. The authors reviewed all such output and take full responsibility for the content of this article.

Appendix A Codebook and operational definitions

Table A1 gives the operational definition, the decision rule and the treatment of ambiguous cases for each indicator. The tracking-cookie taxonomy follows.

Table A1: Operational definitions of the coded indicators.

Indicator	Coded present when	Coded absent or partial when
Privacy notice	A document describing the processing of personal data is reachable from the home page or one click below it	**P** if its stated scope covers only part of the institution's processing; absent if unreachable from the home page
Framework cited	The notice names a specific instrument, for example the LOPDP, the GDPR or the Personal Data (Privacy) Ordinance	Absent for generic formulations such as "applicable law"; **P** for a named framework that does not apply to the institution
Rights listed	At least three of access, rectification, erasure, restriction, objection and portability are enumerated, with a route to exercise them	Absent for an unenumerated mention of "your rights"; **P** if enumerated without any route
Privacy contact or DPO	A named role, a dedicated address or a dedicated form for privacy matters is given	Absent for a general institutional contact form or switchboard number
Accessibility declared	An accessibility statement, an accessibility policy or an accessibility toolbar is published	Absent when accessibility is mentioned only inside another document
Transport security	The home page is served over HTTPS with a chain that a current browser accepts without warning	Absent for expired, self-signed or incomplete chains
Tracking before consent	At least one cookie whose name matches the taxonomy below is present before any interaction with a consent interface	Absent when no matching cookie is present
No level-A failure	axe-core reports no violation carrying a level-A tag	Absent when at least one level-A violation is reported

Tracking-cookie taxonomy. A cookie was classified as belonging to an advertising or analytics vendor when its name matched one of the following patterns, applied as regular expressions to the cookie name. The list was extended after the multi-vantage replication of Section 3.6; the families added at that point are marked with an asterisk. The extended list is applied throughout by reclassifying the stored cookie names, so that no measurement was repeated and every reported figure derives from the same recorded observations. At the site level the two lists differ on these data by a single site, an Ecuadorian institution whose only matching cookies belong to the Sourcebuster family. At the level of individual cookies they differ substantially, which is why the extension was made.

- *Google Analytics and Ads*: `^_ga`, `^_gid$`, `^__utm`, `^_gcl_`, `^_gac_`; *Privacy Sandbox**: `^receive-cookie-deprecation$`.
- *YouTube and DoubleClick*: `^YSC$`, `^VISITOR_INFO`, `^VISITOR_PRIVACY`*, `^__Secure-YNID$`*, `^__Secure-ROLLOUT`*, `^IDE$`, `^test_cookie$`, `^DSID$`*.
- *Meta*: `^_fbp$`, `^_fbc$`, `^fr$`. *TikTok*: `^_tt_`, `^_ttp$`, `^ttcsid`*.
- *Microsoft Clarity, Advertising and UET*: `^_clck$`, `^_clsk$`, `^MUID$`, `^MR$`*, `^SM$`*, `^SRM_B$`*, `^ANONCHK$`*, `^CLID$`*, `^_uet`.
- *LinkedIn Insight Tag**: `^bcookie$`, `^bscookie$`, `^lidc$`, `^li_sugr$`, `^li_gc$`, `^UserMatchHistory$`, `^AnalyticsSyncHistory$`.
- *TapAd**: `^TapAd_`. *StackAdapt**: `^sa-user-id`. *Snapchat**: `^_scid`, `^sc_at$`.
- *X and Twitter*: `^muc_ads$`*, `^personalization_id$`, `^_twpid$`*.
- *Hotjar*: `^_hj`. *Siteimprove*: `^nmstat$`. *Matomo*: `^_pk_`. *PixelYourSite and Sourcebuster*: `^pys`, `^sbjs_`*. *Pinterest*: `^_pin_`.

- *Adobe and Tealium*: `^AMCV_`, `^s_`, `^utag`, `^mbox`; *ContentSquare**: `^_cs_`.
- *Baidu*: `^Hm_lvt_`, `^Hm_lpvt_`, `^HMACCOUNT`, `^BAIDUID$`. *Quantcast*: `^__qca$`. *LiveIntent*: `^_lc2_`, `^ln_or$`.

The list is name-based and therefore conservative: a vendor using an unlisted or randomised cookie name is not counted, which is one reason the figures in Section 4.2 are lower bounds. The consequence of the initial omissions is quantified in Section 4.6: under the original list the fall in vendor cookies at the European vantage point is 33.4%, and under the extended list it is 43.9%.

Appendix B Indicator-by-jurisdiction mapping

Table B1 states, for each jurisdiction in the study, the instrument accepted as the applicable framework and whether the jurisdiction imposes a general duty of prior consent for non-essential cookies. It is the basis for the distinction, maintained throughout Section 4, between observed conduct and legal non-compliance. Figure E2 in Appendix E places these regimes on a common timeline.

Table B1: Applicable framework and prior-consent duty by jurisdiction.

Jurisdiction	Instrument accepted as applicable framework	General prior-consent duty for non-essential cookies
Ecuador	LOPDP and its implementing regulation [5, 53]	No equivalent of Art. 5(3) ePrivacy; consent is required for processing, not as a storage-access rule
European Union	GDPR [21] and national transpositions of the ePrivacy Directive [20]	Yes, under Art. 5(3) ePrivacy as transposed [19]
United Kingdom	United Kingdom GDPR and Data Protection Act 2018 [51]	Yes, under the national ePrivacy transposition
Switzerland	Federal Act on Data Protection (FADP) [58]	No general prior-consent rule; transparency and opt-out apply
United States	California Consumer Privacy Act (CCPA) [8] for California; the Family Educational Rights and Privacy Act (FERPA) [61] and the Health Insurance Portability and Accountability Act (HIPAA) [64] govern education records and protected health information respectively and do not govern the public portal	No general prior-consent duty; opt-out model
Canada	Provincial public-sector statutes for public universities [34, 35, 44]; the Personal Information Protection and Electronic Documents Act (PIPEDA) [49] for commercial activity	No general prior-consent duty; implied consent recognised
China	Personal Information Protection Law (PIPL) [46]	Consent required for non-essential processing
Japan	Act on the Protection of Personal Information (APPI) [26]	No general prior-consent duty for cookies as such
South Korea	Personal Information Protection Act (PIPA) [45]	Consent required for non-essential processing
Singapore	Personal Data Protection Act (PDPA) [50]	Deemed consent model; no general prior-consent duty
Australia	Privacy Act 1988 [48]	No general prior-consent duty
Hong Kong SAR	Personal Data (Privacy) Ordinance (PDPO) [36]	No general prior-consent duty

The consequence for interpretation is direct: in the majority of the jurisdictions studied, setting a tracking cookie before consent is not in itself unlawful. The indicator is retained because it measures the exposure of the visitor, which is the concern of this article, but it is not, and is nowhere presented as, a compliance rate.

Appendix C Documentary matrix, benchmark group

Legend: ✓ present; – absent; **P** partial, meaning limited scope or a generic framework; *NV* not verifiable by this method. Rankings: QS 2026, THE 2026 and ARWU 2025, giving the position, with "–" for absence from that ranking's top 75. Country in ISO 3166-1 alpha-2. Columns: Not. (first-level privacy notice), Frm. (specific legal instrument named), Rts. (data-subject rights enumerated), DPO (privacy contact or data protection officer identified), Ckp. (a dedicated cookie policy is published, a documentary indicator distinct from the live banner measurement of Section 4.2 and reported in aggregate in Section 4.1), Acc. (accessibility statement or tools declared). Coding date: August 2026. Cells marked † were corrected after the manual re-verification described in Section 3.4.

Table C1: Documentary matrix of the 63 benchmark institutions.

#	Abbr.	C	QS	THE	ARWU	Not.	Frm.	Rts.	DPO	Ckp.	Acc.
1	MIT	US	1	2	3	✓	✓	✓	✓	–	✓
2	Stanford	US	3	5	2	✓	✓	✓	✓	✓	✓
3	Harvard	US	5	5	1	✓	–†	–	–	–	✓
4	Oxford	GB	4	1	6	✓	✓	✓	✓	✓	✓
5	Cambridge	GB	6	3	4	✓†	**P**	–	–	✓	✓
6	Caltech	US	10	7	9	✓	✓	–	✓	–	✓
7	UC Berkeley	US	17	9	5	✓	✓	–	✓	–	✓
8	Princeton	US	25	3	7	✓	✓	✓	✓	–	✓
9	Imperial	GB	2	8	26	✓	✓	**P**	–	✓	✓
10	U Chicago	US	13	15	10	✓	**P**	✓	✓	–	–
11	ETH Zurich	CH	7	11	22	✓	✓	✓	✓	–	✓
12	Yale	US	21	10	11	✓	✓	**P**	✓	–	✓
13	UPenn	US	15	14	14	✓	**P**	✓	✓	–	✓
14	UCL	GB	9	22	14	✓	**P**	✓	✓	✓	✓
15	Cornell	US	16	18	12	✓	**P**	✓	✓	–	✓
16	Tsinghua	CN	17	12	18	✓	✓	✓	✓	–	–
17	Peking	CN	14	13	23	**P**	**P**	–	–	–	–
18	Johns Hopkins	US	24	16	19	✓	✓	✓	✓	–	✓
19	Columbia	US	38	20	8	✓	✓	✓	✓	✓	✓
20	Toronto	CA	29	21	25	✓	✓	✓	✓	–	✓
21	UCLA	US	46	18	16	✓	✓	✓	✓	–	✓
22	NUS	SG	8	17	56	✓	✓	✓	✓	–	–
23	U Tokyo	JP	36	26	31	✓	✓	✓	–	–	–
24	TUM	DE	22	27	45	✓	✓	✓	✓	–	–
25	Melbourne	AU	19	37	38	✓	✓	✓	✓	–	–
26	Edinburgh	GB	34	29	37	✓	✓	✓	✓	–	✓
27	EPFL	CH	22	35	44	✓	✓	✓	✓	–	✓
28	Michigan	US	45	23	33	✓	–	–	✓	–	–
29	Northwestern	US	42	30	31	✓	**P**	✓	✓	✓	✓
30	Fudan	CN	30	36	41	–	–	–	–	–	–
31	PSL	FR	28	48	34	✓	✓	✓	✓	–	–
32	HKU	HK	11	33	67	✓	✓	✓	✓	–	✓
33	Zhejiang	CN	49	39	24	✓	**P**	✓	✓	–	–
34	NYU	US	55	31	28	✓	✓	✓	✓	–	✓
35	SJTU	CN	47	40	30	–	–	–	–	–	–
36	KCL	GB	31	38	61	✓	**P**	✓	✓	✓	✓
37	UCSD	US	66	47	20	✓	✓	✓	✓	–	✓
38	LMU	DE	58	34	42	✓	✓	✓	✓	–	✓
39	Duke	US	62	28	46	✓	✓	✓	✓	–	✓
40	Manchester	GB	35	56	46	✓	✓	**P**	✓	–	✓
41	UBC	CA	40	45	53	✓	✓	**P**	✓	–	✓
42	Sydney	AU	25	53	72	✓	✓	✓	✓	–	✓
43	Paris-Saclay	FR	70	68	13	✓	✓	✓	✓	✓	✓
44	Kyoto	JP	57	61	46	✓	**P**	✓	✓	–	–
45	Illinois	US	70	41	53	✓	✓	✓	✓	–	✓
46	UT Austin	US	68	50	49	✓	✓	✓	✓	–	✓
47	UW	US	–	25	17	✓	✓	✓	✓	–	✓
48	NTU	SG	12	31	–	✓	✓	✓	✓	–	–

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(Table C1, continued)

#	Abbr.	C	QS	THE	ARWU	Not.	Frm.	Rts.	DPO	Ckp.	Acc.
49	McGill	CA	27	41	–	✓	✓	✓	✓	✓	✓
50	CUHK	HK	32	41	–	✓	✓	✓	–	–	✓
51	CMU	US	52	24	–	✓	✓	✓	✓	–	✓
52	Wisconsin	US	–	53	36	✓	✓	–	✓	✓	✓
53	USTC	CN	–	51	40	**P**	✓	✓	✓	–	–
54	WUSTL	US	–	67	26	✓	–	–	✓	–	✓
55	Monash	AU	36	58	–	✓	✓	–	–	–	✓
56	SNU	KR	38	58	–	✓	✓	✓	✓	–	–
57	Heidelberg	DE	–	49	51	✓	✓	✓	✓	–	✓
58	HKUST	HK	44	58	–	✓	✓	✓	✓	–	✓
59	Karolinska	SE	–	53	50	✓	**P**	✓	✓	✓	✓
60	TU Delft	NL	47	57	–	✓	✓	✓	✓	✓	✓
61	ANU	AU	32	73	–	✓	✓	**P**	✓	–	✓
62	KU Leuven	BE	60	46	–	✓	✓	✓	✓	✓	✓
63	Queensland	AU	42	–	65	✓	✓	✓	✓	–	✓

Appendix D Documentary matrix, Ecuadorian census

Legend: ✓ present; – absent; **P** partial, meaning limited scope, for example postgraduate programmes, admissions or a mobile application only; NV not verifiable by this method, meaning the site could not be retrieved, the document was not machine-readable, or the element is rendered dynamically and could not be detected. Type: P public, C co-financed, A self-financed. Columns: Notice, LOPDP (the law is named expressly), Rights, DPO, Acc. (accessibility statement or tools declared), Transp. (LOTAIP transparency section published). Institutions created in 2024 or later are marked ‡. Coding date: August 2026. The live cookie measurements are reported in aggregate in Section 4.2 and per site in the deposited dataset; they are not reproduced here, because the documentary and live indicators are different constructs and presenting them in one grid invited the confusion this table avoids.

Table D1: Documentary matrix of the 63 Ecuadorian HEIs.

#	Abbr.	Type	Notice	LOPDP	Rights	DPO	Acc.	Transp.
1	EPN	P	✓	✓	✓	–	–	✓
2	ESPOL	P	✓	✓	✓	✓	–	✓
3	ESPOCH	P	–	–	–	–	–	✓
4	ESPAM MFL	P	–	–	–	–	–	✓
5	ESPE	P	✓	✓	✓	✓	✓	✓
6	UCE	P	*NV*	–	–	–	–	✓
7	UCuenca	P	✓	✓	✓	–	–	✓
8	UG	P	✓	✓	✓	✓	–	✓
9	UNL	P	–	–	–	–	–	✓
10	UAE	P	–	–	–	–	–	✓
11	UTA	P	✓	*NV*	*NV*	*NV*	*NV*	✓
12	UTB	P	✓	✓	✓	✓	–	✓
13	UTC	P	✓	–	–	–	–	*NV*
14	UTMACH	P	✓	✓	✓	✓	–	✓
15	UTM	P	–	–	–	–	–	✓
16	UTN	P	**P**	✓	✓	–	–	✓
17	UTEQ	P	**P**	✓	–	–	–	✓
18	UTELVT	P	–	–	–	–	–	✓
19	UEB	P	✓	*NV*	*NV*	*NV*	✓	✓
20	UNESUM	P	**P**	*NV*	–	–	–	✓
21	UNEMI	P	✓	✓	✓	–	–	✓
22	UEA	P	✓	–	✓	–	–	✓
23	UPSE	P	**P**	✓	✓	✓	**P**	✓
24	ULEAM	P	–	–	–	–	–	✓
25	UNACH	P	✓	✓	✓	✓	✓	✓
26	UNAE	P	–	–	–	–	–	✓
27	UPEC	P	–	**P**	–	–	–	✓
28	Yachay Tech	P	–	–	–	–	–	✓
29	Ikiam	P	✓	✓	✓	✓	–	✓
30	UArtes	P	✓	✓	✓	✓	–	✓

Continued on next page

(Table D1, continued)

#	Abbr.	Type	Notice	LOPDP	Rights	DPO	Acc.	Transp.
31	Amawtay Wasi	P	–	–	–	–	–	✓
32	USECIPOL‡	P	**P**	–	–	–	–	✓
33	IAEN	P	✓	✓	✓	–	✓	✓
34	UASB	P	✓	✓	✓	✓	–	✓
35	FLACSO	P	–	–	–	–	–	✓
36	PUCE	C	✓	✓	✓	✓	–	✓
37	UCSG	C	✓	✓	✓	✓	–	✓
38	UCACUE	C	✓	✓	✓	✓	✓	✓
39	ULVR	C	–	–	–	–	–	✓
40	UTPL	C	✓	✓	✓	✓	–	*NV*
41	UTE	C	✓	–	–	–	✓	*NV*
42	UDA	C	–	–	–	–	–	✓
43	UPS	C	✓	✓	✓	–	–	✓
44	UISEK	A	✓	–	✓	–	–	–
45	UEES	A	✓	✓	✓	✓	–	–
46	USFQ	A	✓	✓	✓	✓	–	–
47	UDLA	A	✓	✓	✓	✓	–	–
48	UIDE	A	✓	✓	✓	✓	–	✓
49	UNIANDES	A	✓	–	✓	–	–	–
50	UPACÍFICO	A	*NV*	–	–	–	–	–
51	UTI	A	✓	–	✓	–	–	✓
52	Casa Grande	A	✓	✓	✓	✓	–	–
53	UTEG	A	✓	✓	✓	✓	–	✓
54	UISRAEL	A	✓	*NV*	*NV*	*NV*	*NV*	*NV*
55	UDET	A	–	–	–	–	–	–
56	UMET	A	✓	*NV*	*NV*	*NV*	–	✓
57	U. Otavalo	A	*NV*	*NV*	–	–	–	*NV*
58	USGP	A	✓	✓	✓	✓	–	✓
59	U. Hemisferios	A	✓	✓	✓	✓	–	✓
60	UNIB.E	A	✓	–	✓	–	–	–
61	ECOTEC	A	✓	✓	✓	✓	–	✓
62	UDR	A	–	–	–	–	–	–
63	UBE	A	✓	✓	✓	✓	✓	–

Appendix E Supplementary figures

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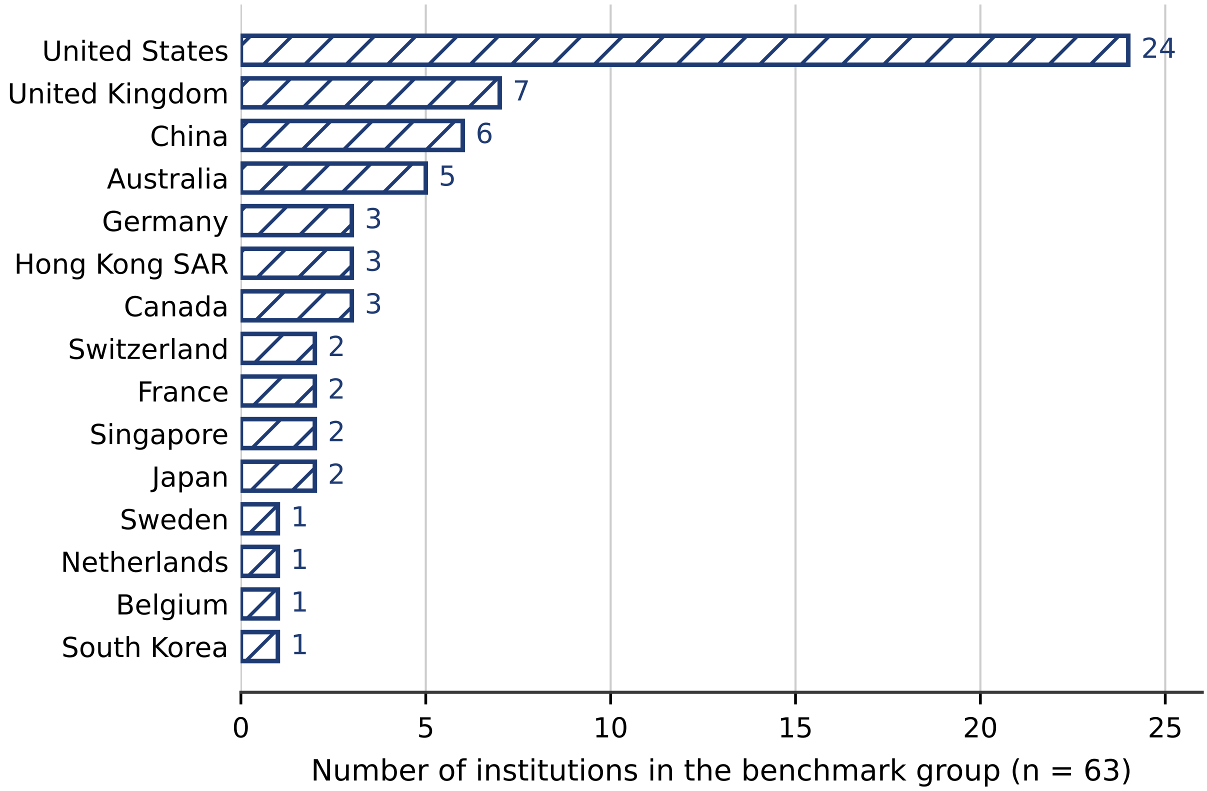

Fig. E1 Composition of the benchmark group by country. The group is the extreme upper tail of three international rankings and is not a sample of the world's universities; see Section 3.1.

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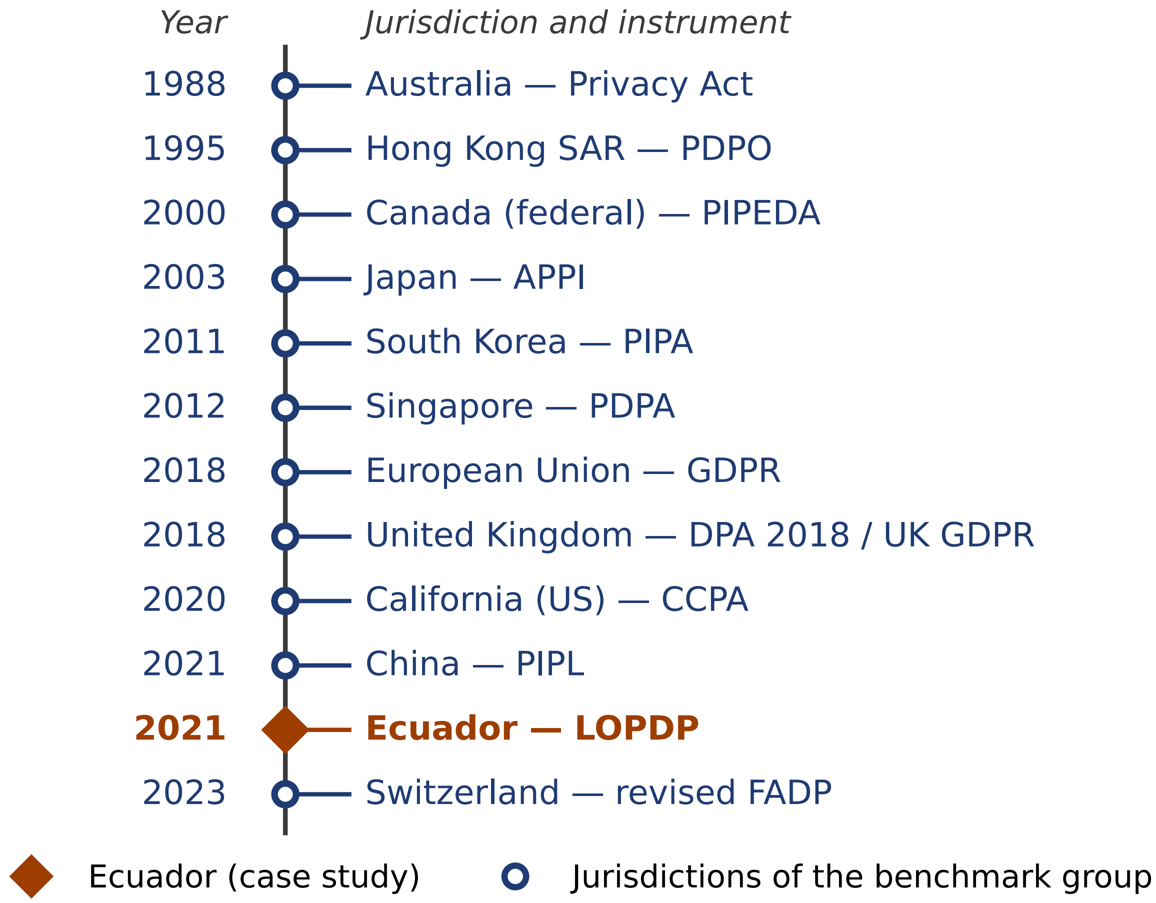

Fig. E2 Year in which the current general data-protection regime took effect in each jurisdiction of the study. For the United States, which has no comprehensive federal statute, California's CCPA is used. Ecuador belongs to the most recent wave, alongside China.

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